TECHNICAL DISCUSSION MEMO

SORh as a Two-Parameter Generalization of SOR

CFL roles, optimal relaxation, nonlinear propagation,
and compatible discrete operators

Central interpretation *Classical SOR is the subfamily in which directional correction propagation and residual injection have the same normalized strength. SORh separates those roles through ω and CFL; alternative left- and right-hand-side discretizations preserve the classical reduction only when their causal matrix splittings match.*

Scope. This memo consolidates the derivations and conclusions developed while reviewing the one- and two-dimensional SORh and SORp formulations, their finite-CFL behavior, observed nonlinear performance, and extensions to higher-band discrete operators.

Topic	Result
SORp CFL	Uniformly scales the classical SOR correction; only $\omega_{\mathrm{P}}\mathrm{CFL}_{\mathrm{P}}$ matters in the linear constant-coefficient case.
SORh CFL	Scales residual injection but not the causal correction propagator.
Classical slice	For the default centered scheme, $\mathrm{CFL}_{\mathrm{H}} = 1$ and $\alpha = 2\omega_{\mathrm{H}}$ give classical lexicographic SOR.
Operator compatibility	Exact reduction requires the LHS increment stencil to match $D^{-1}L$ of the RHS elliptic operator.

Prepared July 30, 2026

Numerical Solver Research Notes

Contents

1 Executive summary **1**

2 Discrete setting and notation **1**

3 SORp and SORh correction equations **1**

3.1 Parabolic SOR 1

3.2 Hyperbolic SOR 2

4 Classical SOR as a subfamily **2**

5 Exact finite-CFL linear spectrum **2**

5.1 From the point update to the iteration matrix 3

5.2 What the affine identity says about the spectrum 3

5.3 Why the optimal ω_{H} usually stays classical 3

5.4 What CFL changes at the optimum 4

6 Stability and the fixed-residual tradeoff **4**

7 Why nonlinear coupling changes the comparison **5**

8 The one-dimensional and two-dimensional mapping **5**

9 Compatibility of discrete operators **6**

9.1 Three useful levels of compatibility 6

10 Higher-order spatial BDF on the LHS **6**

11 Larger Laplacian stencils **7**

11.1 Coefficient matching example 7

11.2 Nine-point stencil 7

12 An operator-first generalized SOR design **7**

13 Recommended numerical program **8**

14 Conclusions **8**

A Compact formula sheet **9**

1 Executive summary

- **SORp is classical SOR with a rescaled parameter.** For the linear constant-coefficient problem, $\alpha = \omega_{\mathrm{P}}\mathrm{CFL}_{\mathrm{P}}$ exactly. Changing $\mathrm{CFL}_{\mathrm{P}}$ can be offset by changing ω_{P} inversely.
- **SORh has two effective strengths.** Its normalized propagation strength is $\alpha_{\mathrm{prop}} = 2\omega_{\mathrm{H}}$, while its residual-injection strength is $\alpha_{\mathrm{res}} = 2\omega_{\mathrm{H}}\mathrm{CFL}_{\mathrm{H}}$.
- **The classical reduction is exact at** $\mathrm{CFL}_{\mathrm{H}} = 1$**.** Then $\alpha_{\mathrm{prop}} = \alpha_{\mathrm{res}}$, and the default BDF1/centered SORh sweep is classical lexicographic SOR.
- **The finite-CFL linear spectrum is an affine image of the** $\mathrm{CFL} = 1$ **spectrum.** For the model scheme, $G(r, \omega) = (1 - r)I + rG(1, \omega)$. For the modeled grids with $N \geq 6$, the optimal ω remains the classical formula for $0 < r \leq 1$, while the best convergence factor worsens as r decreases.
- **Nonlinear performance is explained by decoupling.** SORh propagates an accepted upstream correction independently of the local diffusion coefficient, while SORp attenuates that propagation by a local coefficient ratio.
- **Discretization compatibility is algebraic.** Higher-order BDF or larger Laplacian stencils retain the same classical slice only if their causal supports, coefficient ratios, diagonal normalization, ordering, and boundary closures agree.

Practical conclusion *For the standard linear Poisson solve, CFL near one and the classical ω_{H} formula are the natural optimum. In nonlinear, anisotropic, few-sweep, or multigrid settings, a lower CFL can be useful because it damps new residual forcing while preserving strong causal propagation.*

2 Discrete setting and notation

Consider a homogeneous error equation for a discrete Poisson problem. Let u^k be the old iterate, let $\delta = u^{k+1} - u^k$ be the correction, and let R^k denote the unnormalized old-iterate stencil residual. In one dimension,

$$R_j^k = u_{j-1}^k - 2u_j^k + u_{j+1}^k. \tag{1}$$

The two pseudo-time scalings are dimensionally different. Their normalized stability numbers are

$$r_{\mathrm{H}} = \frac{\widetilde{v}\,\Delta t_{\mathrm{H}}}{\Delta x} = \mathrm{CFL}_{\mathrm{H}}, \qquad r_{\mathrm{P}} = \frac{\widetilde{v}\,\Delta t_{\mathrm{P}}}{\Delta x^2} = \frac{\mathrm{CFL}_{\mathrm{P}}}{2}. \tag{2}$$

Thus the same numerical CFL does not represent the same pseudo-time step: Δt_{H} scales like Δx, whereas Δt_{P} scales like Δx^2. In the stationary iteration, however, the important question is where each normalized number appears in the correction equation.

3 SORp and SORh correction equations

3.1 Parabolic SOR

The relaxed parabolic Gauss–Seidel update uses the newly computed left neighbor:

$$u_j^{k+1} = u_j^k + \omega_{\mathrm{P}} r_{\mathrm{P}} \left(u_{j-1}^{k+1} - 2u_j^k + u_{j+1}^k \right). \tag{3}$$

Writing the new left value as $u_{j-1}^k + \delta_{j-1}$ gives

$$\delta_j = \beta \left(\delta_{j-1} + R_j^k \right), \qquad \beta = \omega_{\mathrm{P}} r_{\mathrm{P}} = \frac{\omega_{\mathrm{P}}\mathrm{CFL}_{\mathrm{P}}}{2}. \tag{4}$$

Both propagation and residual injection have the same coefficient β. Consequently, ω_P and CFL_P are redundant in the linear fixed-coefficient iteration.

3.2 Hyperbolic SOR

Sign correction to the report excerpt *The finite-r_H expanded update must contain $+\omega_H(r_H - 1)u^k_{j-1}$. The printed minus sign conflicts with the later iteration-matrix equation and would fail to preserve the Poisson fixed point when $r_H \neq 1$.*

With the corrected sign, subtracting the old iterate and collecting the old-level Poisson residual gives

$$\delta_j = \omega_H \delta_{j-1} + \omega_H r_H R^k_j. \tag{5}$$

The propagation coefficient is ω_H, while the residual coefficient is $\omega_H r_H$. CFL therefore changes the balance between the two mechanisms instead of uniformly rescaling the whole SOR correction.

Role	SORp coefficient	SORh coefficient
Upstream correction propagation	$\omega_P \mathrm{CFL}_P/2$	ω_H
Old residual injection	$\omega_P \mathrm{CFL}_P/2$	$\omega_H \mathrm{CFL}_H$
Independent linear controls	One effective product	Two effective strengths

4 Classical SOR as a subfamily

Classical one-dimensional SOR has correction coefficient $\alpha/2$. Hence SORp satisfies

$$\alpha = 2\omega_P r_P = \omega_P \mathrm{CFL}_P. \tag{6}$$

This mapping is exact for every parabolic CFL in the linear constant-coefficient scheme. The classical optimum therefore gives

$$\omega_{P,\mathrm{opt}} = \frac{2}{\mathrm{CFL}_P\,[1 + \sin(\pi/N)]}. \tag{7}$$

For SORh, define two classical-normalized parameters:

$$\alpha_{\mathrm{prop}} = 2\omega_H, \qquad \alpha_{\mathrm{res}} = 2\omega_H \mathrm{CFL}_H. \tag{8}$$

At $\mathrm{CFL}_H = 1$ these agree, and the SORh correction becomes classical SOR with $\alpha = 2\omega_H$. The resulting optimum is

$$\omega_{H,\mathrm{opt}} = \frac{1}{1 + \sin(\pi/N)}. \tag{9}$$

Interpretation *Classical SOR is the diagonal $\alpha_{\mathrm{prop}} = \alpha_{\mathrm{res}}$ in a two-parameter plane. Finite-CFL SORh moves below that diagonal: it preserves propagation while reducing residual injection.*

5 Exact finite-CFL linear spectrum

Let $e^k = u^k - u^*$. Define $G_H(r_H, \omega_H)$ by $e^{k+1} = G_H(r_H, \omega_H)e^k$. Thus G_H is the one-sweep error-propagation matrix, and $\rho(G_H)$ is the asymptotic error-reduction factor per sweep; it is not the Poisson operator. The point recurrence is

$$\delta^k_j = \omega_H \delta^k_{j-1} + \omega_H r_H R^k_j, \tag{10}$$

where $r_H = \mathrm{CFL}_H$ in the normalized scheme.

5.1 From the point update to the iteration matrix

The one-dimensional recurrence is used first because it exposes the algebra with one upstream neighbor. The square-grid two-dimensional derivation is identical after replacing S by $(S_x + S_y)/2$ and using the normalized five-point residual.

Symbol	Meaning
$e^k = u^k - u^*$	Error after sweep k
$\delta^k = u^{k+1} - u^k$	Correction accepted during that sweep
$R^k = f - Au^k = -Ae^k$	Old-iterate Poisson residual
S	Strict lower-shift operator, $(Sx)_j = x_{j-1}$
$G_{\mathrm{H}}(r_{\mathrm{H}}, \omega_{\mathrm{H}})$	One-sweep matrix defined by $e^{k+1} = G_{\mathrm{H}} e^k$

The point recurrence is lower triangular because the current-sweep correction δ^k_{j-1} is already available. Collecting all points gives

$$(I - \omega_{\mathrm{H}} S)\, \delta^k = \omega_{\mathrm{H}} r_{\mathrm{H}} R^k. \tag{5.1}$$

Solving this triangular system makes the two roles visible. The inverse $(I - \omega_{\mathrm{H}} S)^{-1}$ repeatedly transports an accepted correction downstream; r_{H} multiplies the residual only after that propagator has been fixed:

$$\delta^k = \omega_{\mathrm{H}} r_{\mathrm{H}}\, (I - \omega_{\mathrm{H}} S)^{-1}\, R^k. \tag{5.2}$$

Since $R^k = -Ae^k$, define the full-CFL correction operator $B_{\mathrm{H}}(\omega_{\mathrm{H}})$ and substitute into $e^{k+1} = e^k + \delta^k$:

$$B_{\mathrm{H}}(\omega_{\mathrm{H}}) = \omega_{\mathrm{H}}\, (I - \omega_{\mathrm{H}} S)^{-1}\, A, \qquad G_{\mathrm{H}}(r_{\mathrm{H}}, \omega_{\mathrm{H}}) = I - r_{\mathrm{H}} B_{\mathrm{H}}(\omega_{\mathrm{H}}). \tag{5.3}$$

At $r_{\mathrm{H}} = 1$, $G_{\mathrm{H}}(1, \omega_{\mathrm{H}}) = I - B_{\mathrm{H}}$. Eliminating B_{H} yields the exact finite-CFL identity

$$G_{\mathrm{H}}(r_{\mathrm{H}}, \omega_{\mathrm{H}}) = (1 - r_{\mathrm{H}}) I + r_{\mathrm{H}} G_{\mathrm{H}}(1, \omega_{\mathrm{H}}). \tag{5.4}$$

What equation (5.4) proves *CFL does not alter the triangular propagation operator inside a sweep. It under-relaxes the complete full-CFL correction by mixing the identity step (no progress) with the* CFL = 1 *sweep. This is the precise algebra behind "unchanged propagation, reduced residual injection."*

5.2 What the affine identity says about the spectrum

Equation (5.4) is an affine polynomial in the CFL = 1 matrix. Therefore the two matrices have the same eigenvectors. If $G_{\mathrm{H}}(1, \omega_{\mathrm{H}})\xi_i = \lambda_i(1, \omega_{\mathrm{H}})\xi_i$, then

$$\lambda_i(r_{\mathrm{H}}, \omega_{\mathrm{H}}) = 1 - r_{\mathrm{H}} + r_{\mathrm{H}} \lambda_i(1, \omega_{\mathrm{H}}). \tag{5.5}$$

Changing CFL does not create new spatial modes: the eigenvectors are unchanged. Reducing CFL also does not move the eigenvalues toward zero; it pulls them toward 1, so each sweep removes less of the error.

5.3 Why the optimal ω_{H} usually stays classical

At CFL = 1, set $\alpha = 2\omega_{\mathrm{H}}$. For a consistently ordered Poisson matrix, a classical-SOR eigenvalue λ associated with Jacobi eigenvalue μ satisfies Young's relation

$$(\lambda + \alpha - 1)^2 = \alpha^2 \mu^2 \lambda, \qquad \mu = \rho_J = \cos(\pi/N). \tag{5.6}$$

The two dominant real roots merge when the discriminant in (5.6) vanishes. Solving that condition gives the classical transition point

$$\alpha^* = \frac{2}{1+\sqrt{1-\mu^2}} = \frac{2}{1+\sin(\pi/N)}, \qquad \omega_{\mathrm{H,opt}} = \frac{\alpha^*}{2}. \tag{5.7}$$

Below α^*, the dominant positive root decreases as α increases. Above α^*, the roots are a complex pair. Applying the affine map (5.5) to that pair gives

$$\frac{d\,|1-r+r\lambda|^2}{d\alpha} = 2r\left\{r(\alpha-1)+(1-r)(\alpha\mu^2-1)\right\}. \tag{5.8}$$

Scope of the unchanged-optimum statement *For the 1D and square-grid 2D Poisson spectra with $N \geq 6$, $\alpha^*\mu^2 \geq 1$, so (5.8) is nonnegative for $\alpha \geq \alpha^*$ and every $0 < r_{\mathrm{H}} \leq 1$. The left branch falls and the right branch rises, fixing the minimum at (5.7). On $N < 6$ grids with very small CFL, the optimum can shift upward slightly even though (5.4) remains exact.*

5.4 What CFL changes at the optimum

$$\rho_{\mathrm{opt}}(r_{\mathrm{H}}) = 1 - r_{\mathrm{H}} + r_{\mathrm{H}}\rho_{\mathrm{opt}}(1) = 1 - 2r_{\mathrm{H}}\left(1-\omega_{\mathrm{H,opt}}\right). \tag{5.9}$$

Equation (5.9) shows that lowering $\mathrm{CFL_H}$ moves the best contraction factor toward 1 while leaving $\omega_{\mathrm{H,opt}}$ fixed for $N \geq 6$. This explains why experiments at CFL = 0.8 and 0.9 continue to favor essentially the same ω_{H} formula, even though the derivation by direct comparison with classical SOR was performed at CFL = 1.

6 Stability and the fixed-residual tradeoff

Keeping $\omega_{\mathrm{H}} r_{\mathrm{H}}$ fixed preserves the coefficient on the old residual, but it does not preserve the iteration matrix. Increasing ω_{H} strengthens every downstream reuse of an accepted correction. Repeated substitution produces

$$\delta_j = sR_j + qsR_{j-1} + q^2 sR_{j-2} + \cdots, \qquad q = \omega_{\mathrm{H}}, \quad s = \omega_{\mathrm{H}} r_{\mathrm{H}}. \tag{11}$$

Thus q is a per-cell propagation gain. Values above one amplify rather than attenuate corrections along a sweep and become unstable on sufficiently large grids.

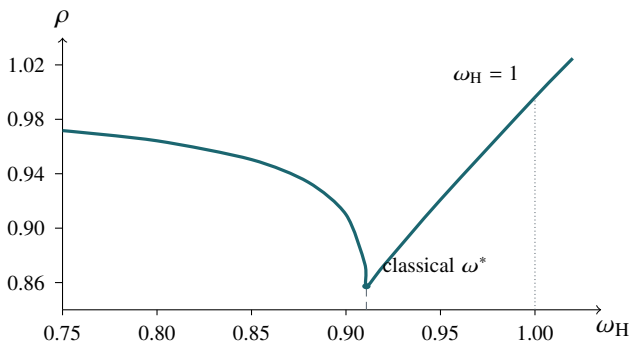


Figure 1: Exact one-dimensional spectral radius along a constant residual-product path, $\omega_{\mathrm{H}} r_{\mathrm{H}} = 0.7286$ with $N = 32$. The sharp minimum remains near the classical propagation parameter; increasing ω_{H} toward or above one rapidly destroys convergence.

ω_H	CFL_H	$\omega_H CFL_H$	$\rho(G)$	Interpretation
0.911	0.800	0.729	0.857	Near the optimum
1.002	0.727	0.729	0.999	Nearly stagnant
1.821	0.400	0.729	3.334	Strongly divergent

A substantially lower CFL can still be meaningful, but the useful combination is normally ω_H close to, and below, one with reduced residual injection. An inverse scaling $\omega_H \propto 1/CFL_H$ incorrectly treats ω_H as if it were the redundant outer multiplier in SORp.

7 Why nonlinear coupling changes the comparison

Let $F_j(u)$ be a variable-coefficient discrete diffusion residual using face coefficients $v_{j-1/2}$ and $v_{j+1/2}$. Freezing those coefficients during a local linearization gives the approximate SORp correction

$$\delta_j^P \approx \frac{\omega_P CFL}{2v_{max}} \left[F_j(u^k) + v_{j-1/2}\delta_{j-1}\right]. \tag{12}$$

The corresponding SORh update is

$$\delta_j^H = \omega_H \delta_{j-1} + \frac{\omega_H CFL}{v_{max}} F_j(u^k). \tag{13}$$

SORp therefore attenuates propagation by the local ratio $v_{j-1/2}/v_{max}$, whereas SORh does not. No single ω_P can make all local propagation strengths optimal when the coefficients vary strongly. Raising ω_P enough to compensate in weak-coupling regions can destabilize strong-coupling regions.

Local CFL interpretation *A global CFL of* 0.8 *may correspond to a local residual strength of only* 0.16 *where* $v_{face}/v_{max} = 0.2$*, while the SORh propagation coefficient remains near one. The nonlinear solver is therefore already operating in a strong-propagation, weak-injection regime.*

N	SORp sweeps	SORh sweeps	Sweep ratio
17	122	104	1.17
33	448	210	2.13
65	1,741	419	4.16
129	6,709	815	8.23
257	25,478	1,584	16.08

Table 1: Stored one-dimensional nonlinear benchmark results at CFL = 0.8. The growing ratio is consistent with uniform directional correction transport being the dominant advantage of SORh.

8 The one-dimensional and two-dimensional mapping

On a square grid with the centered five-point Laplacian, the default two-dimensional SORh correction is

$$\delta_{ij} = \frac{\omega_H}{2}\left(\delta_{i-1,j} + \delta_{i,j-1}\right) + \frac{\omega_H CFL}{2} R_{ij}^k. \tag{14}$$

Classical lexicographic SOR for the five-point operator uses $\alpha/4$ on the two lower-neighbor corrections and on the unnormalized residual. At CFL = 1, $\alpha = 2\omega_H$ again matches every term. The prefactor $c_1 = 1/(h_x + h_y)$

supplies the normalization that compensates for the change from a diagonal value of 2 in one dimension to 4 in two dimensions.

Implementation caveat *The two-dimensional examples in this repository configure SORp with lexicographic ordering, matching the default SORh directional sweep. Selecting red–black ordering instead changes the lower-triangular splitting and therefore the iteration matrix, even when the scalar coefficients correspond.*

9 Compatibility of discrete operators

Let the chosen RHS elliptic operator be split according to the actual sweep ordering as $A = D - L - U$. Classical SOR in correction form is

$$\left(I - \alpha D^{-1} L\right) \delta = \alpha D^{-1}(f - Au). \tag{15}$$

A general SORh discretization can be written

$$(I - \omega P)\delta = \omega\gamma Q(f - Au), \tag{16}$$

where P is the causal increment-propagation operator, Q is the residual normalization, and γ contains CFL. An exact classical-SOR slice exists if and only if there is a scalar α satisfying

$$\omega P = \alpha D^{-1} L, \qquad \omega\gamma Q = \alpha D^{-1}. \tag{17}$$

Compatibility test	Required match
Causal support	P uses every and only already visited neighbor represented in L.
Coefficient ratios	Propagation weights match the normalized lower-stencil weights.
Diagonal normalization	Q matches D^{-1} up to the same scalar mapping.
Ordering	The same lexicographic, directional, red–black, or block ordering defines L.
Boundary closure	Startup and boundary rows preserve the same split, not only the interior.

9.1 Three useful levels of compatibility

- **Fixed-point compatibility:** $\delta = 0$ implies $f - Au = 0$. The method solves the desired discrete equation if it converges.
- **Structural compatibility:** P resembles the lower graph of A but has different weights or additional causal neighbors. The method is a generalized triangular preconditioner.
- **Exact classical compatibility:** P and Q satisfy the two matrix identities above. Equal propagation and residual strengths recover classical SOR.

10 Higher-order spatial BDF on the LHS

For an interior square-grid cell, the current spatial BDF2 implementation yields

$$\delta_{ij} = \omega \left[\frac{2}{3} \left(\delta_{i-1,j} + \delta_{i,j-1} \right) - \frac{1}{6} \left(\delta_{i-2,j} + \delta_{i,j-2} \right) + \frac{\mathrm{CFL}}{3} R_{ij} \right]. \tag{18}$$

The five-point Laplacian has no second-upstream entries in L. No scalar α can match the immediate-neighbor term, remove the second-neighbor term, and match the residual coefficient simultaneously. Consequently, BDF2 plus the five-point Laplacian has no classical-SOR point, including at CFL = 1.

For the BDF blend parameter η, the interior formula is

$$\delta_{ij} = \omega \left[\frac{1+\eta}{2+\eta} \left(\delta_{i-1,j} + \delta_{i,j-1} \right) - \frac{\eta}{2(2+\eta)} \left(\delta_{i-2,j} + \delta_{i,j-2} \right) + \frac{\mathrm{CFL}}{2+\eta} R_{ij} \right]. \tag{19}$$

Only $\eta = 0$, the BDF1 endpoint, removes the second-neighbor term and restores the classical reduction. The startup row and column also fall back to lower-order formulas, making the global preconditioner spatially varying.

Accuracy versus solver compatibility *A higher-order approximation of the mixed derivative is not automatically a better approximation to the lower triangular factor of the elliptic operator. Formal PDE order and preconditioner quality are distinct design objectives.*

11 Larger Laplacian stencils

A larger RHS stencil can be made exactly compatible, but the LHS must propagate all lower-triangular neighbors with matching coefficient ratios. Merely using a wider or higher-order formula on both sides is insufficient.

RHS operator	Lower neighbors in a row-major sweep	Current LHS status
Centered five-point	left; bottom	Exactly compatible with BDF1 at CFL = 1
Wide stride-two	left-2; bottom-2	Not compatible with BDF1 or current BDF2
Isotropic nine-point	left; bottom-left; bottom; bottom-right	Missing diagonal increment propagation
Fourth-order cross stencil	left; left-2; bottom; bottom-2	Same offsets as BDF2, but different ratios
Operator-derived causal LHS	Exactly the support of $D^{-1}L$	Compatible by construction

11.1 Coefficient matching example

The standard fourth-order centered second derivative has lower-neighbor ratio $16:-1$, while BDF2 propagation has ratio $4:-1$. The supports overlap, but no scalar normalization can reconcile the ratios. A custom causal LHS could use $16:-1$, but it would not be standard BDF2 and its first-derivative moment conditions would need to be rechecked.

11.2 Nine-point stencil

For the isotropic nine-point operator, the lower triangular set depends on ordering. With complete previous rows, bottom-left and bottom-right are already available along with left and bottom. Exact SOR compatibility therefore requires diagonal correction propagation with the Laplacian's axis-to-diagonal ratio of $4:1$. The current LHS includes only left and bottom, so it defines a partial Gauss–Seidel preconditioner.

12 An operator-first generalized SOR design

If the stationary solver interpretation is primary, a compatible two-parameter method can be constructed directly for any discrete operator:

$$\left(I - \alpha_{\mathrm{prop}} D^{-1} L \right) \delta = \alpha_{\mathrm{res}} D^{-1} (f - Au). \tag{20}$$

- Choose A, the boundary closure, and the sweep ordering first.
- Form D and the actual lower triangular part L.
- Use $D^{-1}L$ as the causal propagation operator.
- Set $\alpha_{\text{prop}} = \alpha_{\text{res}}$ to recover classical SOR.
- Use $\alpha_{\text{res}}/\alpha_{\text{prop}}$ as the H-like residual-to-propagation ratio, analogous to CFL.
- Only afterward ask whether the algebraic propagator has a useful pseudo-time PDE interpretation.

If the hyperbolic PDE discretization is primary instead, an independently selected BDF or directional LHS remains legitimate. It should then be treated as a new triangular preconditioner whose optimal parameters and stability region must be derived from its own iteration matrix.

13 Recommended numerical program

Experiment	Purpose
Matched linear baseline	Verify one-sweep equality for lexicographic SORp and BDF1 SORh at CFL = 1 and $\omega_{\text{P}} = 2\omega_{\text{H}}$.
Two-parameter maps	Sweep $q = \alpha_{\text{prop}}$ and $s = \alpha_{\text{res}}$ rather than raw ω and CFL; plot spectral radius and nonlinear iteration counts.
BDF blend spectra	Build $G(\eta, \omega, \text{CFL})$ and determine how the second-upstream term changes the stability boundary.
Stencil-matched propagators	Construct $D^{-1}L$ propagators for the wide, nine-point, and fourth-order operators; compare with the current PDE-derived LHS.
Nonlinear local diagnostics	Record distributions of CFL $v_{\text{face}}/v_{\max}$ and correlate low local coupling with SORp propagation slowdown.
Multigrid smoothing study	Optimize high-frequency damping separately from solve-to-convergence spectral radius.

Suggested parameter coordinates *For default SORh, report $q = \omega_{\text{H}}$ and $s = \omega_{\text{H}}\text{CFL}_{\text{H}}$ in one dimension, or total directional propagation $q = q_x + q_y = \omega_{\text{H}}$ together with residual strength s in two dimensions. These coordinates expose the mechanism directly.*

14 Conclusions

The revised interpretation removes a false dichotomy between parabolic and hyperbolic SOR. On the linear centered Poisson problem, both are related to the same classical triangular splitting. SORp keeps propagation and residual injection locked together; SORh exposes them as separate controls.

The directional terms are not an additional acceleration mechanism at CFL = 1: they are the mechanism by which Gauss–Seidel already propagates corrections during a sweep. Their distinctive value appears away from the classical slice, especially when nonlinear or variable coefficients would otherwise attenuate correction transport.

This interpretation also clarifies discretization design. The exact classical reduction is not guaranteed by using a higher-order LHS and a higher-order RHS. It is guaranteed by matching the causal increment operator to the diagonal-normalized lower triangular part of the chosen elliptic stencil. BDF2 and larger stencils remain useful possibilities, but when that matching fails they define new preconditioners and require independent spectral analysis.

One-sentence summary *SORh is best understood as a transport-preconditioned, two-strength SOR family; the current BDF1/centered scheme contains classical SOR exactly, while other discretization pairs do so only when their matrix splittings are compatible.*

A Compact formula sheet

Concept	Formula
SORp correction	$\delta_j = \dfrac{\omega_{\mathrm{P}}\mathrm{CFL}_{\mathrm{P}}}{2}(\delta_{j-1} + R_j)$
SORh correction	$\delta_j = \omega_{\mathrm{H}}\delta_{j-1} + \omega_{\mathrm{H}}\mathrm{CFL}_{\mathrm{H}}R_j$
Classical normalized roles	$\alpha_{\mathrm{prop}} = 2\omega_{\mathrm{H}}, \quad \alpha_{\mathrm{res}} = 2\omega_{\mathrm{H}}\mathrm{CFL}_{\mathrm{H}}$
SORp mapping	$\alpha = \omega_{\mathrm{P}}\mathrm{CFL}_{\mathrm{P}}$
Default SORh classical slice	$\mathrm{CFL}_{\mathrm{H}} = 1, \quad \alpha = 2\omega_{\mathrm{H}}$
Optimal SORh propagation	$\omega_{\mathrm{H,opt}} = \dfrac{1}{1 + \sin(\pi/N)}$
Finite-CFL spectrum	$G(r, \omega) = (1 - r)I + rG(1, \omega)$
General compatibility	$\omega P = \alpha D^{-1}L, \quad \omega\gamma Q = \alpha D^{-1}$
Operator-first family	$(I - \alpha_{\mathrm{prop}}D^{-1}L)\delta = \alpha_{\mathrm{res}}D^{-1}(f - Au)$